

Standard tracking tells you where players move.

TF15 tells you how they move and where they look.

The Bundesliga TRACAB TF15 system captures 141 million data points per match, full 3D skeleton at 50 fps. Today it powers only automated event detection. The body intelligence it contains goes unmeasured.

AWI, AWARENESS INDEX

Cognitive scanning rate Discrete head rotation events per minute

3D Keypoint Extraction Derived from nose, neck, and ear
Impossible with 2D tracking

Direct Measurement A scan is a head rotation of 45 degrees or more within 0.5 seconds. One sustained turn equals one event.

PQI, PRESSURE QUALITY INDEX

Pressing Mechanics Score 0 to 100 during genuine press frames
(10+ frames within 5m of ball)

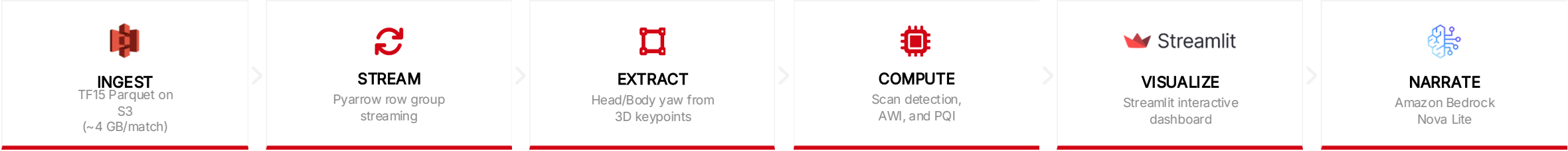
The Formula $PQI = 0.40 \times \text{orientation} + 0.30 \times \text{stance} + 0.30 \times \text{proximity}$

Biomechanical Optimum Stance optimum is 130 degree knee flexion for maximum explosive capacity

*A player can scan brilliantly but press with poor mechanics (high AWI, low PQI), or press perfectly without pre-scanning (low AWI, high PQI).
Elite players score high on both.*

Pipeline: S3, Skeleton, Angles, Metrics, Dashboard, AI Narratives

Details the technical pipeline from S3 data to Amazon Bedrock narratives. Key engineering decisions include circular smoothing and SageMaker parallel processing for rapid match analysis.



⚙️ ENGINEERING DECISIONS

DECISION	WHY IT MATTERS
Circular smoothing via sin/cos decomposition	Standard rolling mean breaks at plus/minus 180 degrees. This preserves angular continuity for head and body orientation.
Leading-edge counting	Ensures 1 sustained rotation equals 1 event, rather than counting every frame of the turn. Essential for accurate AWI calculation.
45 degree threshold (not 30 degrees)	XY-plane projection compresses 3D angles. Tuned on Kimmich to filter out micro-adjustments and focus on genuine scanning.
Press frame filter: 10+ consecutive frames	Excludes incidental proximity. Captures genuine pressing intent within 5m of ball carrier for PQI scoring.
SageMaker Processing: 10 parallel jobs	Processes 5 matches across 2 metrics in ~15 min. Reduces local runtime from ~2 hours, enabling rapid iteration.

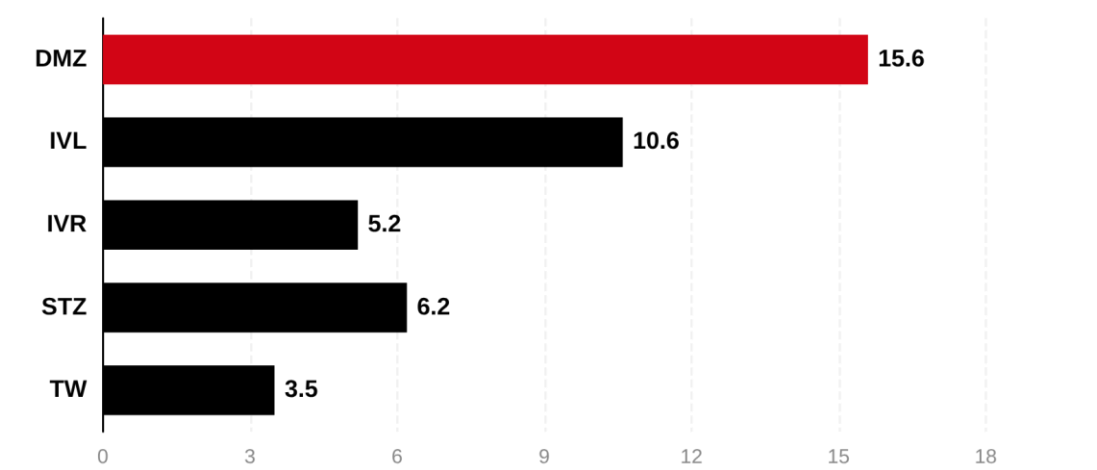
Defensive midfielders are the most cognitively active players on the pitch. **AWI spikes 59% before a pass.**

🏆 TOP 8 AWI LEADERBOARD

RANK	PLAYER	POS	MATCH	AWI
1	Oscar Hojlund	DMZ	SGE-FCB H1	26.90
2	Hugo Larsson	DMZ	SGE-FCU H1	26.26
3	Joshua Kimmich	DMR	FCU-FCB H1	23.38
4	Rani Khedira	DMR	SGE-FCU H1	22.95
5	Oscar Hojlund	DMZ	SGE-FCB H2	22.74
6	Rani Khedira	DMR	SGE-FCU H2	22.27
7	Aljoscha Kemlein	DMZ	FCU-FCB H2	21.89
8	Joshua Kimmich	DMR	FCB-HSV H1	21.77

Cross-half stability $r = 0.660$ (n = 79).AWI is a stable player trait, not match noise.

📊 POSITION AVERAGES (SCANS/MIN)

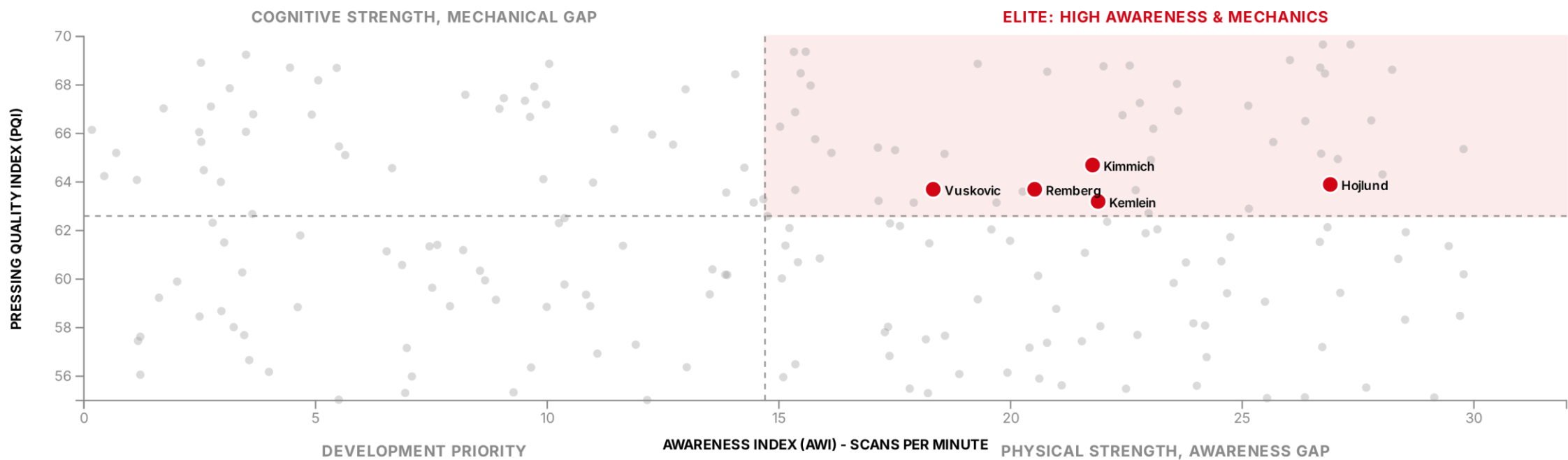


⚡ Pre-Pass Signal

Pre-pass AWI is +59% above full-phase baseline. Players ramp scanning in the 5 seconds before releasing the ball. AWI measures the pre-decision cognitive window.

Scanning awareness and pressing mechanics are independent, not interchangeable.

AWI vs PQI Pearson $r = -0.11$ ($p = 0.12$, $n = 198$). Knowing a player's scan rate tells you almost nothing about their pressing mechanics. Both metrics are needed to identify elite talent.



★ ELITE QUADRANT PERFORMERS

Hojlund (DMZ)	AWI 26.90 PQI 63.9	Kimmich (DMR)	AWI 21.77 PQI 64.7
Kemlein (DMZ)	AWI 21.89 PQI 63.2	Remberg (DMR)	AWI 20.52 PQI 63.7
Vuskovic (IVZ)	AWI 18.33 PQI 63.7		



Strategic Insight: Vuskovic is a centre-back whose scanning profile matches a defensive midfielder. The data flags his tactical versatility and potential for role transition before traditional scouting metrics would detect it.

From 141 million data points per match to **two numbers every coach understands.**



For Clubs

Scouting. Rank by AWI and PQI independently. Find dual-elite players or target specific development gaps.

Fatigue detection. Kimmich AWI drops to 11.29 in FCU-FCB 2nd half. Invisible to GPS.

Positional outliers. Vuskovic (CB) in the elite quadrant flags tactical versatility automatically.



For Broadcasters and Fans

Live overlay. "Hojlund scanned 27 times before that interception."

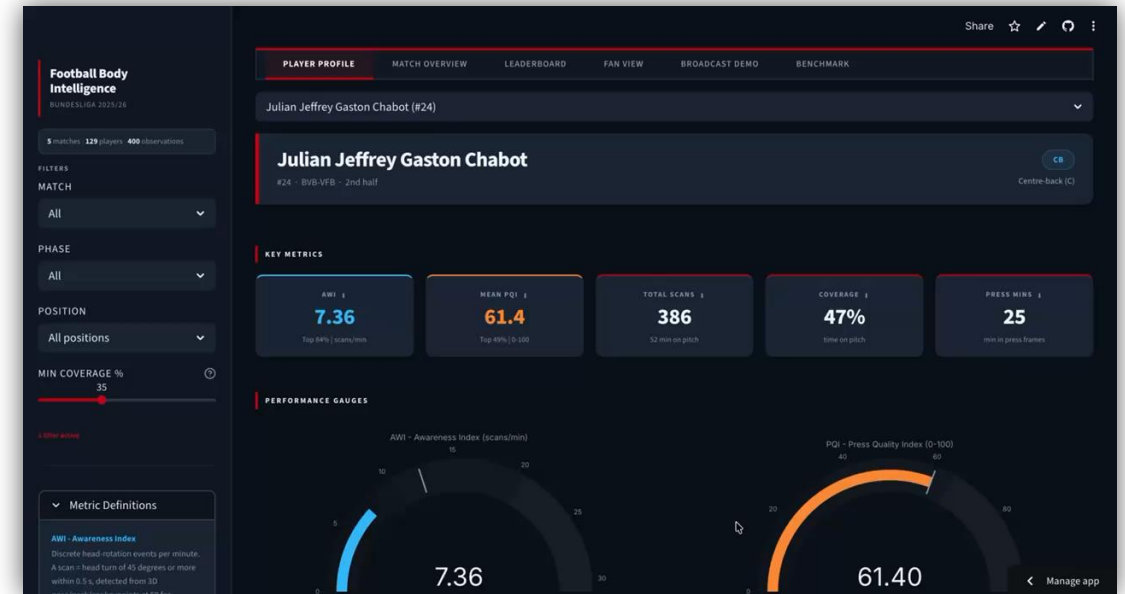
Fan View tab. Broadcast-style top-3 counter, quadrant classification, Body Intelligence leaderboard.



For the DFL

Matchday Grade Metrics. First metrics derived solely from TF15 skeleton data.

Market Edge. Differentiates the 3D product from any 2D competitor.



+59%

The pre-pass AWI spike (~19.2 scans/min) matches the aviation elite level (19 scans/min, Lounis et al., 2021, PLOS ONE). TF15 makes this measurement possible at matchday scale for the first time in team sports.